

Demonstration of Proposed Features

Date	02 July 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Demonstration of Proposed Features

This document summarizes the key features proposed for the OptiCrop project and their implementation status. Each feature was developed, tested, and demonstrated successfully during the project. The system helps users make informed farming decisions by analyzing soil nutrients and environmental conditions using a machine learning model.

S.No	Feature Name	Description	Status	Demonstrated (Yes/No)	Remarks
1	Crop Recommendation	Recommends the most suitable crop using soil nutrients and environmental parameters.	Implemented	Yes	Working successfully.
2	Crop Suitability Analysis	Evaluates how suitable a selected crop is for the given soil and climate conditions.	Implemented	Yes	Displays suitability percentage and status.
3	Top 3 Crop Predictions	Displays the three best crop recommendations with confidence scores.	Implemented	Yes	Helps compare multiple crop options.
4	Parameter Gap Analysis	Identifies which soil or climate parameters are above or below the ideal range.	Implemented	Yes	Provides detailed gap analysis.
5	Farming Tips	Shows practical farming recommendations based on the predicted crop.	Implemented	Yes	Gives useful cultivation advice.

6	Confidence Score	Displays the confidence level of the machine learning prediction.	Implemented	Yes	Helps users understand prediction reliability.
7	Quick Fill Presets	Allows users to quickly load sample values for crops such as Rice, Wheat, Maize, Coffee, and Cotton.	Implemented	Yes	Makes testing and demonstration easier.
8	Responsive Web Interface	Provides a clean and responsive interface for entering inputs and viewing prediction results across devices.	Implemented	Yes	Successfully deployed and tested on the web.

Feature Implementation Summary:

Item	Value
Total Features Proposed	8
Total Features Implemented	8
Total Features Demonstrated	8